Heart Disease Prediction

Helping Doctors Save Lives

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Introduction

Context

Cardiovascular diseases are the **leading cause of death worldwide**.

17.9 million deaths/year

32% of all global deaths

Early detection is crucial

Objectives

- Analyze risk factors
- **Perform statistical tests**
- **Build manual protocol**
- Develop Decision Tree
- **Compare approaches**

The Dataset

270	14	0	0
Patients	Variables	Missing Values	Duplicates

Key Variables

- Age, Sex, Chest pain type
- Blood Pressure (BP), Cholesterol
- Max HR, FBS over 120, EKG results
- Exercise angina, ST depression
- Slope of ST, Thallium, Vessels fluro

Target: Heart Disease

150 healthy / 120 sick patients

Note: Dataset is relatively balanced (44.4% positive class)

Variable Types

Variable	Type
Age	Continuous
Sex	Categorical
Chest pain type	Categorical
BP	Continuous
Cholesterol	Continuous
FBS over 120	Categorical
EKG results	Categorical

Variable	Type
Max HR	Continuous
Exercise angina	Categorical
ST depression	Continuous
Slope of ST	Categorical
Number of vessels	Categorical
Thallium	Categorical

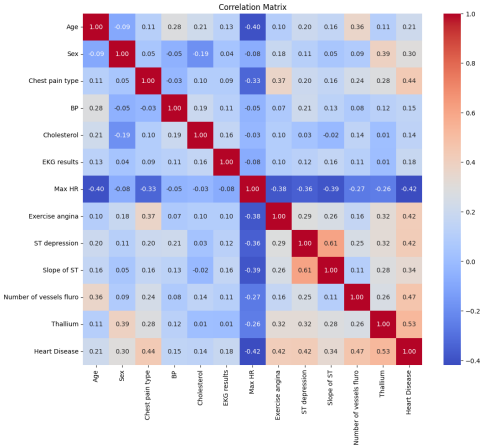
Continuous variables → t-test | **Categorical variables** → Chi^2 test

Descriptive Statistics

Variable	Mean	Median	Q1	Q3	Variance
Age	54.43	55.00	48.00	61.00	82.98
Sex	0.68	1.00	0.00	1.00	0.22
Chest pain type	3.17	3.00	3.00	4.00	0.90
BP	131.34	130.00	120.00	140.00	319.04
Cholesterol	249.66	245.00	213.00	280.00	2671.47
Max HR	149.68	153.50	133.00	166.00	536.65
ST depression	1.05	0.80	0.00	1.60	1.31
Thallium	4.70	3.00	3.00	7.00	3.77

Key observations: Mean age is 54 years, 68% male patients, average BP slightly elevated (131 mmHg)

Correlation Analysis



Positive Correlations:

- Chest pain type (0.44)
- Number of vessels (0.47)
- Thallium (0.53)
- ST depression (0.42)

Negative Correlation:

- Max HR (-0.42)

Statistical tests needed to confirm significance

Statistical Tests Results

P-values by Variable (sorted by significance)

Variable	Test	p-value	Result
Thallium	Chi^2	<0.00001	**SIG**
Chest pain type	Chi^2	<0.00001	**SIG**
Number of vessels	Chi^2	<0.00001	**SIG**
Max HR	t-test	<0.00001	**SIG**
ST depression	t-test	<0.00001	**SIG**
Exercise angina	Chi^2	<0.00001	SIG
Slope of ST	Chi^2	<0.00001	SIG
Sex	Chi^2	<0.00001	SIG
Age	t-test	0.00035	SIG
EKG results	Chi^2	0.01122	SIG
BP	t-test	0.01553	SIG
Cholesterol	t-test	0.02114	SIG
FBS over 120	Chi^2	0.7986	**NS**

✔ **SIGNIFICANT ($p < 0.05$)**

12 out of 13 variables are significantly associated with heart disease

✘ **NOT SIGNIFICANT**

FBS over 120 ($p = 0.80$)
→ **Excluded from protocol**

⚠ **TOP 5 PREDICTORS**

1. Thallium
2. Chest pain type
3. Number of vessels
4. Max HR
5. ST depression

Manual Diagnostic Protocol

RULE 1: HIGH RISK
Chest pain $= 4$ AND Vessels ≥ 1
$\rightarrow$ **PRESENCE (93%)**

RULE 2: MODERATE
Thallium $= 7$ AND ST depression > 1
$\rightarrow$ **Further tests**

RULE 3: MODERATE
Male AND Max HR < 140 AND Age > 55
$\rightarrow$ **Further tests**

RULE 4: LOW RISK
Chest pain ≤ 3 AND Max HR > 150 AND Vessels $= 0$
$\rightarrow$ **ABSENCE (85%)**

Steps: 1) Chest pain $\rightarrow$ 2) Vessels $\rightarrow$ 3) Thallium $\rightarrow$ 4) Max HR $\rightarrow$ 5) ST depression

FBS over 120 excluded ($p = 0.80$, not significant) | Rules based on Chi^2 and t-test results

Modeling: Decision Tree

Why this model?

Interpretable

Understandable rules

Robust

No normalization needed

Visualizable

Clear tree structure

`max_depth=4` | `min_samples_leaf=25` | `class_weight=balanced`

Train/Test Split: 70% / 30% (stratified)
189 training / 81 test patients

Note: FBS over 120 excluded ($p = 0.80$)

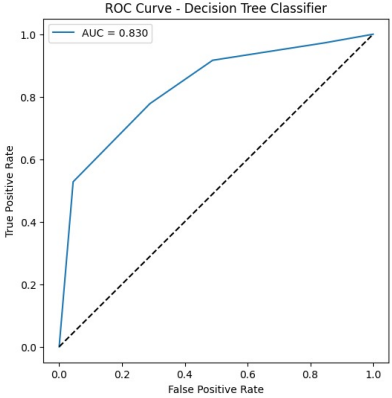
Model Results

Performance

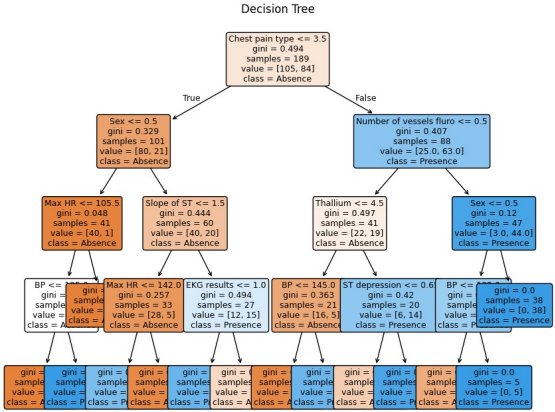
0.830
AUC-ROC

74.1%
Accuracy

Sensitivity: 77.8%

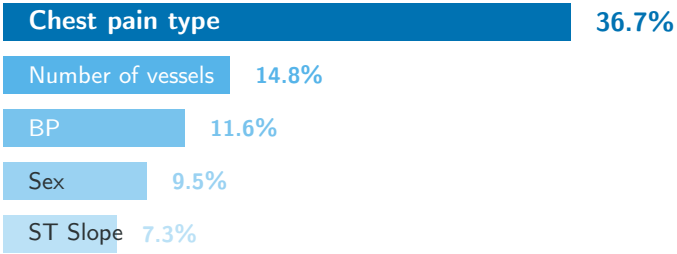


Decision Tree Visualization



Root node: Chest pain type ≤ 3.5 | **Key paths:** Chest pain = 4 + Vessels > 0 + Male $\rightarrow$ Presence

Most Important Variables



Chest pain type is the most discriminant factor (36.7%) - consistent with statistical tests

Comparison: Manual Protocol vs Decision Tree

Manual Protocol

Characteristics:

- 4 simple rules from p-values
- Uses top 5 significant variables
- Sequential evaluation
- Easy to memorize

✓ Simple, no computation
✗ Fixed thresholds, ∼68% accuracy

VS

Decision Tree

Characteristics:

- 15+ learned rules from data
- Optimal thresholds via Gini
- Captures interactions
- Still interpretable (depth=4)

✓ Higher accuracy (74.1%)
✗ Requires computation

RECOMMENDATION: Use Decision Tree for diagnosis (+6% accuracy), Manual Protocol as quick clinical checklist

Conclusion

Results

- **12/13 variables** significantly associated
- Manual protocol from **statistical tests**
- AUC of **0.830**, Accuracy **74.1%**
- Sensitivity of **77.8%**
- **Interpretable** model for doctors
- Top predictor: **Chest pain type**

Perspectives

- Increase dataset size ($n > 1000$)
- Test Random Forest
- External validation
- Clinical integration

Limitation: Small dataset (270 patients). Validate on larger cohort before deployment.

Thank You!

Questions?